Exercise 1: Snowflake Setup (warehouse, database, role, user)

Objective

Create all the Snowflake objects that the rest of the exercises will use: a warehouse for compute, a database and schema to hold data, and a role + user that dbt will use in Exercise 4.

Prerequisites

- Exercise 0 completed (trial active, logged in to Snowsight, account identifier noted down)
- You are logged in as `ACCOUNTADMIN` (default on trial accounts)

Step 1: Open a worksheet

1. In Snowsight, click the **+** icon (top of the page) and select **SQL File**.
2. In the top-right of the worksheet, check:
   - **Role**: `ACCOUNTADMIN` (if not, click and change it)

You will paste each SQL block below and click **Run** (or `Cmd+Return` / `Ctrl+Return`).

Step 2: Create a warehouse

A **Virtual Warehouse** is a cluster of compute resources that executes queries. This one is sized for development:

```
CREATE WAREHOUSE IF NOT EXISTS WH_DBT_DEV
  WAREHOUSE_SIZE = 'X-SMALL'
  AUTO_SUSPEND = 60
  AUTO_RESUME = TRUE
  COMMENT = 'Development warehouse for dbt exercises';
```

What each setting does:

- `WAREHOUSE_SIZE = 'X-SMALL'` : 1 compute node, 1 credit/hour (~$3/hour)
- `AUTO_SUSPEND = 60` : stop billing after 60 seconds with no queries
- `AUTO_RESUME = TRUE` : auto-start when a query arrives

Verify:

```
SHOW WAREHOUSES;
```

You should see your new `WH_DBT_DEV` plus the default `COMPUTE_WH`.

Step 3: Create a database and schema

```sql
CREATE DATABASE IF NOT EXISTS AIRBNB
  COMMENT = 'Airbnb Barcelona dataset for dbt exercises';

CREATE SCHEMA IF NOT EXISTS AIRBNB.DEV
  COMMENT = 'Development schema';
```

Verify in the left panel: **Data > Databases > AIRBNB > DEV** should now exist.

Step 4: Create the dbt role

A **role** is a named collection of privileges. In Exercise 4, dbt will use a dedicated role so it only has the permissions it needs (principle of least privilege).

```sql
CREATE ROLE IF NOT EXISTS TRANSFORM_ROLE
  COMMENT = 'Role used by dbt to transform data';
```

Grant the role access to the warehouse and database:

```sql
-- Warehouse permissions
GRANT USAGE ON WAREHOUSE WH_DBT_DEV TO ROLE TRANSFORM_ROLE;

-- Database permissions
GRANT USAGE ON DATABASE AIRBNB TO ROLE TRANSFORM_ROLE;
GRANT CREATE SCHEMA ON DATABASE AIRBNB TO ROLE TRANSFORM_ROLE;
GRANT ALL ON SCHEMA AIRBNB.DEV TO ROLE TRANSFORM_ROLE;
GRANT ALL ON FUTURE TABLES IN SCHEMA AIRBNB.DEV TO ROLE TRANSFORM_ROLE;
GRANT ALL ON FUTURE VIEWS IN SCHEMA AIRBNB.DEV TO ROLE TRANSFORM_ROLE;
```

> ***Why `FUTURE`?*** *Without this, dbt could create new tables/views but would not automatically have privileges on them. The `FUTURE` grants ensure that any object dbt creates in the schema is accessible to the role.*

Step 5: Create the dbt user

```sql
CREATE USER IF NOT EXISTS dbt_user
  PASSWORD = 'changeme'
  DEFAULT_ROLE = TRANSFORM_ROLE
  DEFAULT_WAREHOUSE = WH_DBT_DEV
  DEFAULT_NAMESPACE = AIRBNB.DEV
  MUST_CHANGE_PASSWORD = FALSE
  COMMENT = 'Service user for dbt connections';

GRANT ROLE TRANSFORM_ROLE TO USER dbt_user;
```

> ***Important****: the password is `changeme`. Write it down — Exercise 4 will need it.*
>
> *In a real production setup you would use a strong password or (better) key-pair authentication. For this classroom exercise `changeme` is fine.*

Step 6: Set your active context

Still as `ACCOUNTADMIN`, tell the worksheet which warehouse, database and schema to use by default:

```
USE WAREHOUSE WH_DBT_DEV;
USE DATABASE AIRBNB;
USE SCHEMA DEV;
```

Verify:

```
SELECT CURRENT_WAREHOUSE(), CURRENT_DATABASE(), CURRENT_SCHEMA(), CURRENT_ROLE();
```

Expected:

CURRENT_WAREHOUSE()	CURRENT_DATABASE()	CURRENT_SCHEMA()	CURRENT_ROLE()
WH_DBT_DEV	AIRBNB	DEV	ACCOUNTADMIN

Step 7: Final verification

Run one last check to confirm everything is in place:

```
-- Objects created
SHOW WAREHOUSES LIKE 'WH_DBT_DEV';
SHOW DATABASES LIKE 'AIRBNB';
SHOW ROLES LIKE 'TRANSFORM_ROLE';
SHOW USERS LIKE 'DBT_USER';
```

Each of the four `SHOW` statements should return exactly one row.

Checklist

- [ ] `WH_DBT_DEV` warehouse exists (X-Small, auto-suspend 60s)
- [ ] `AIRBNB.DEV` database/schema exists
- [ ] `TRANSFORM_ROLE` role exists with privileges on the warehouse + schema
- [ ] `dbt_user` user exists with password `changeme` and role `TRANSFORM_ROLE`
- [ ] Your worksheet context is set to `WH_DBT_DEV` / `AIRBNB.DEV`

Key Concepts

Concept	What it means
Virtual Warehouse	Independent compute cluster that executes queries
Auto-suspend	Stop billing after N seconds of inactivity
Database > Schema > Table	Snowflake's three-level naming hierarchy

Role	Named collection of privileges (RBAC)
User	Identity that connects to Snowflake; can have multiple roles
GRANT ... FUTURE	Automatically grant privileges on objects created later

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Questions

1. Why does Snowflake separate warehouses (compute) from databases (storage)?
2. If you set `AUTO_SUSPEND = 60` and nobody queries for 2 minutes, what happens? What happens when the next query arrives?
3. Why create a dedicated `dbt_user` instead of using your own `ACCOUNTADMIN` account for dbt?